OVERVIEW OF ENERGY DATA ANALYTICS

ENERGY DATA ANALYTICS

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01 | Data in Energy Companies

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Data is the New Infrastructure of the Energy Transition

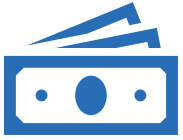
Energy companies collect millions of data points every second.
But most decisions are still made from spreadsheets.

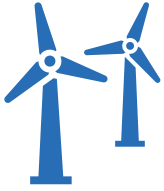
Mastering data is no longer optional
it’s survival

Why the stakes are high?

*The global big data analytics in energy market was valued at USD **9.51 billion** in 2021 and is projected to reach USD **36.76 billion** by 2031*

Allied Market Research, 2025





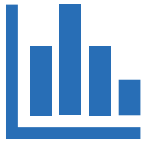
Renewables will make up ~42% of electricity mix by 2028

IEA, World Energy Outlook 2023

Fossil fuels still account for 86% of global energy supply

Energy Institute Statistical Review, 2024

The Role of Data in the Energy Transition



Optimization

New Opportunity

Strategic Asset

Energy company ecosystem

Renewable Generation

	Origination	Development	Monitoring
Data	Geospatial Meteorological Grid	Engineering & Design Financial & Regulatory Construction Data	Operational data (scada) Weather Maintenance data
Use Case	Site selection Resource Assestment Competitive Benchmarking	Detailed Plant Design & Optimization Risk Assessment & Financial Modeling	Performance Optimization Fault Detection Regulatory Reporting

Trading & Energy procurement

	Market Analysis	Execution	Risk Management
Data	Market Price Data Fundamentals Weather Forecasts	Order Book Exchange/Broker APIs Financial & Regulatory Construction Data	Portfolio Position Data Credit Ratings Counterparty Data
Use Case	Price Forecasting Opportunity Identification	Automated Trading & Optimal Execution Algorithmic Strategy Backtesting	Market & Credit Risk Regulatory Compliance Reporting Stress Testing

Retail Utility

	Customer Value	Portfolio & Pricing	Operation
Data	Customer Profile Credit & Payment Data Market & Competitor Pricing	Load Profile Marketing campaign Market & Competitor	Interaction Field Service Network Event
Use Case	Churn Prediction Proactive Communication Debt Management	Load Forecasting long Customer base evolution Targeted Product Bundling	Outage Management Call Center Optimization Field Workforce Dispatch

Breaking the Walls: Data Unification

Overcoming Data Silos

Isolated data limits visibility and hampers quick decision-making, creating significant barriers to organizational agility.

Integrated Analytics for Insights

Combining different data sources enables deeper analytics, leading to smarter and more holistic business strategies.

Governance and Standards

Establishing clear standards and governance ensures consistent, trusted data across the organization.

Ensuring Interoperability

Technology solutions should enable seamless data flow across different platforms and teams for maximal value.

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Why it’s important to remove silos
Example on predictive maintenance

02 | Building a Data-Driven Organization

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Building an Enterprise Data Ecosystem

1. Data

Collecting & Contextualizing

Clean, integrated, and governed Data Assets.

2. Information

Analyzing & Interpreting

Business Metrics, Reports, and Insights (The 'What').

3. Knowledge

Modeling & Predicting

Predictive Models (e.g., Churn, Price), Foresight (The 'Why' / 'What if').

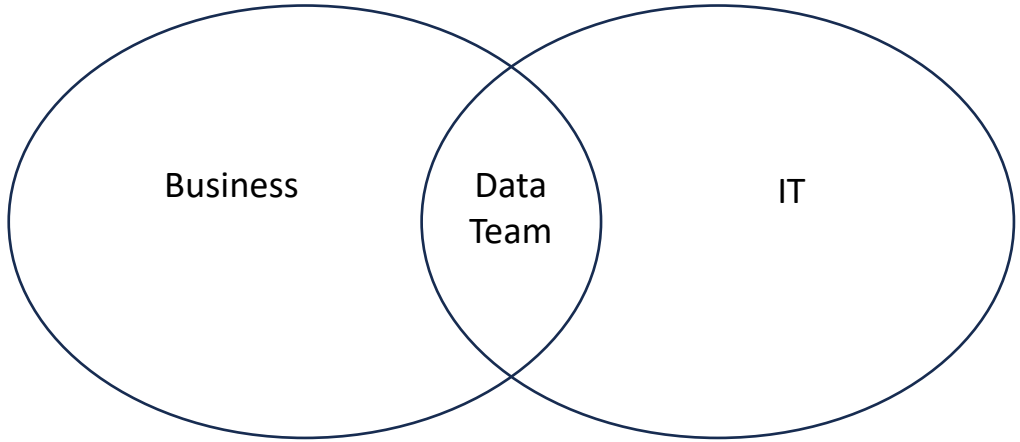
4. Intelligence

Actioning & Automating

Automated Decisions, Optimized Processes, and New Revenue.

Governance, Standardization and Federated Architecture

Data Teams
A translator between Technology and Business



Data Team

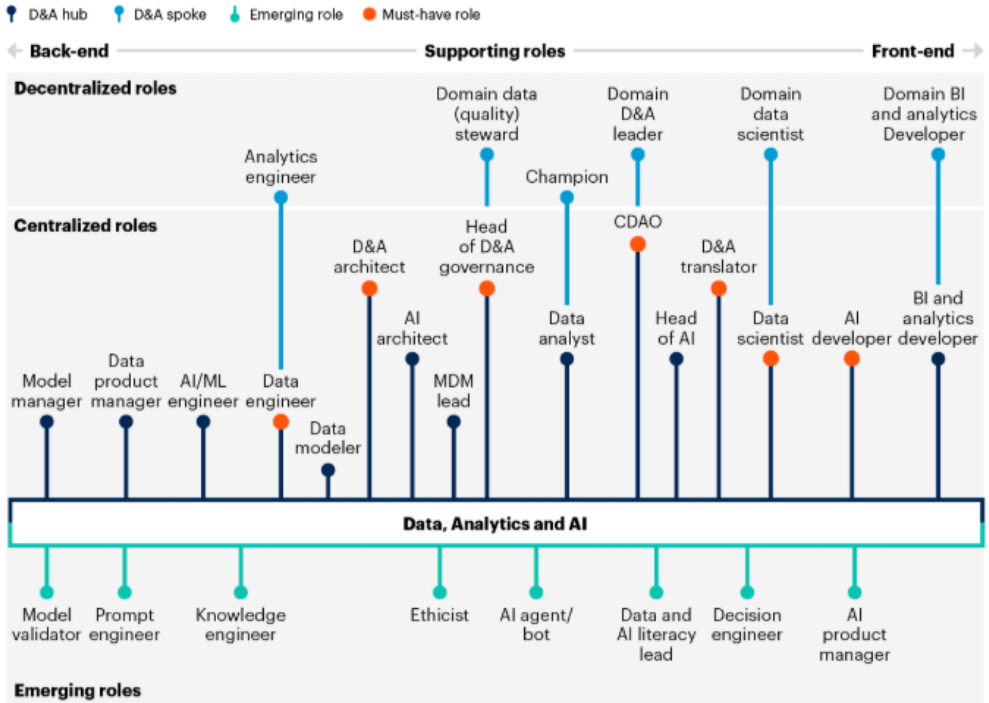
Data Analyst

Data Scientist

Data Engineer

Data Governance Specialist

Must-Have Roles in Data, Analytics and AI



Source: Gartner
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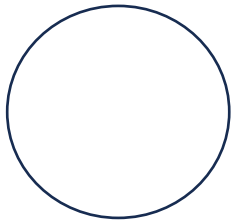
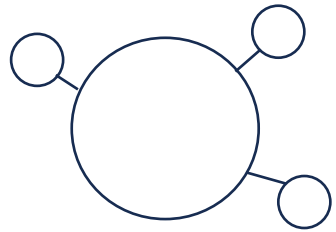
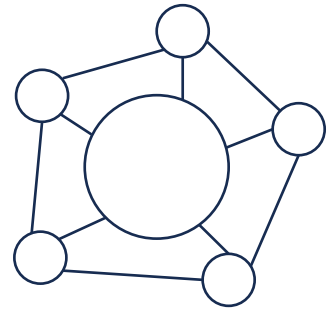
Scaling the Organization



Data Hub



Data Hub +
Business Spoke



Federated /
Data Mesh

culture *communication* *widespread ownership* Maturity

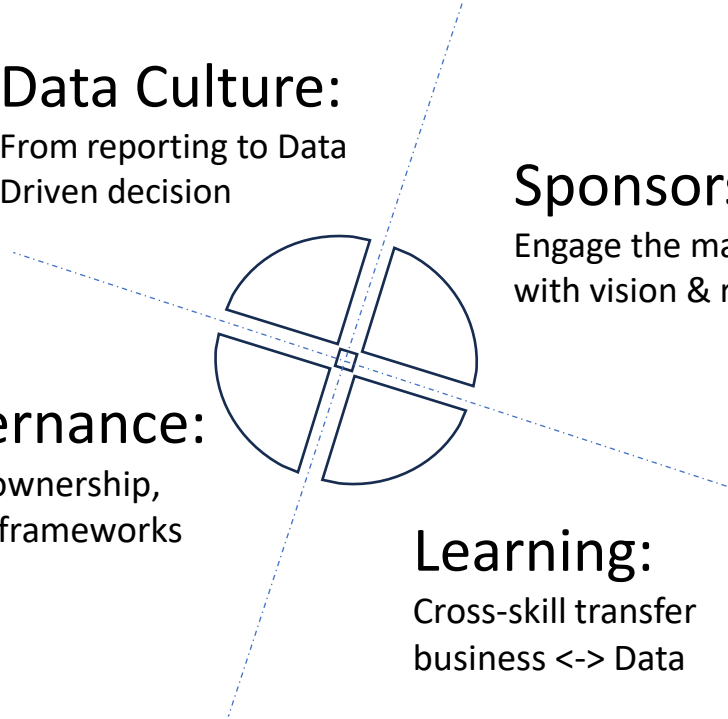
Change Management Challenges

Data Culture:

From reporting to Data Driven decision

Sponsorship:

Engage the management with vision & results



Governance:

Define ownership, tools & frameworks

Learning:

Cross-skill transfer business <-> Data

03 | Data Platform & Enabling Technologies

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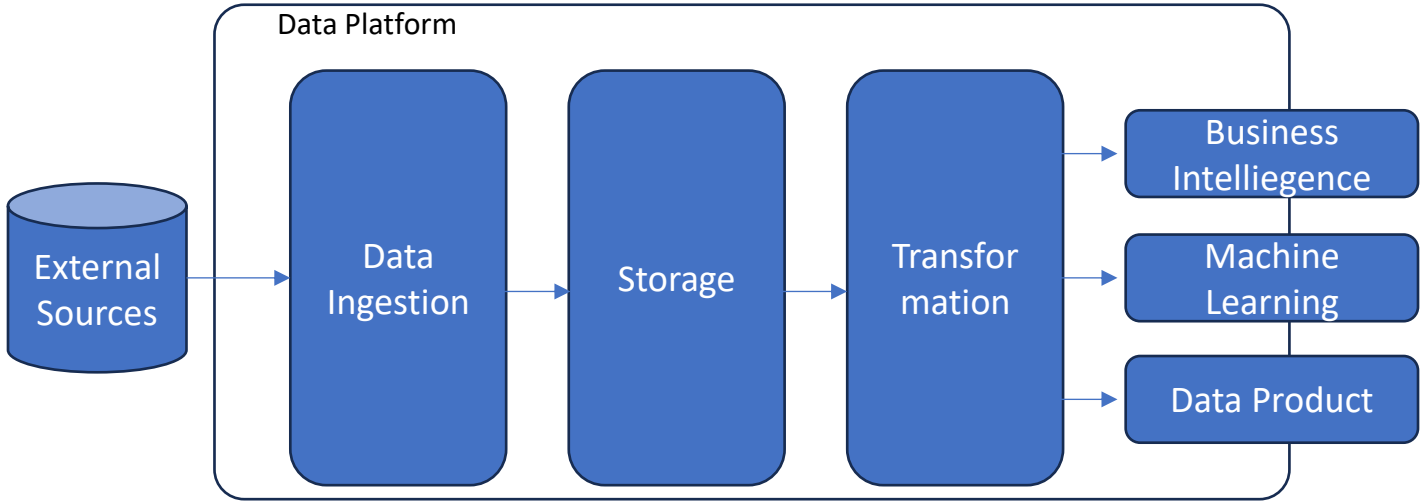
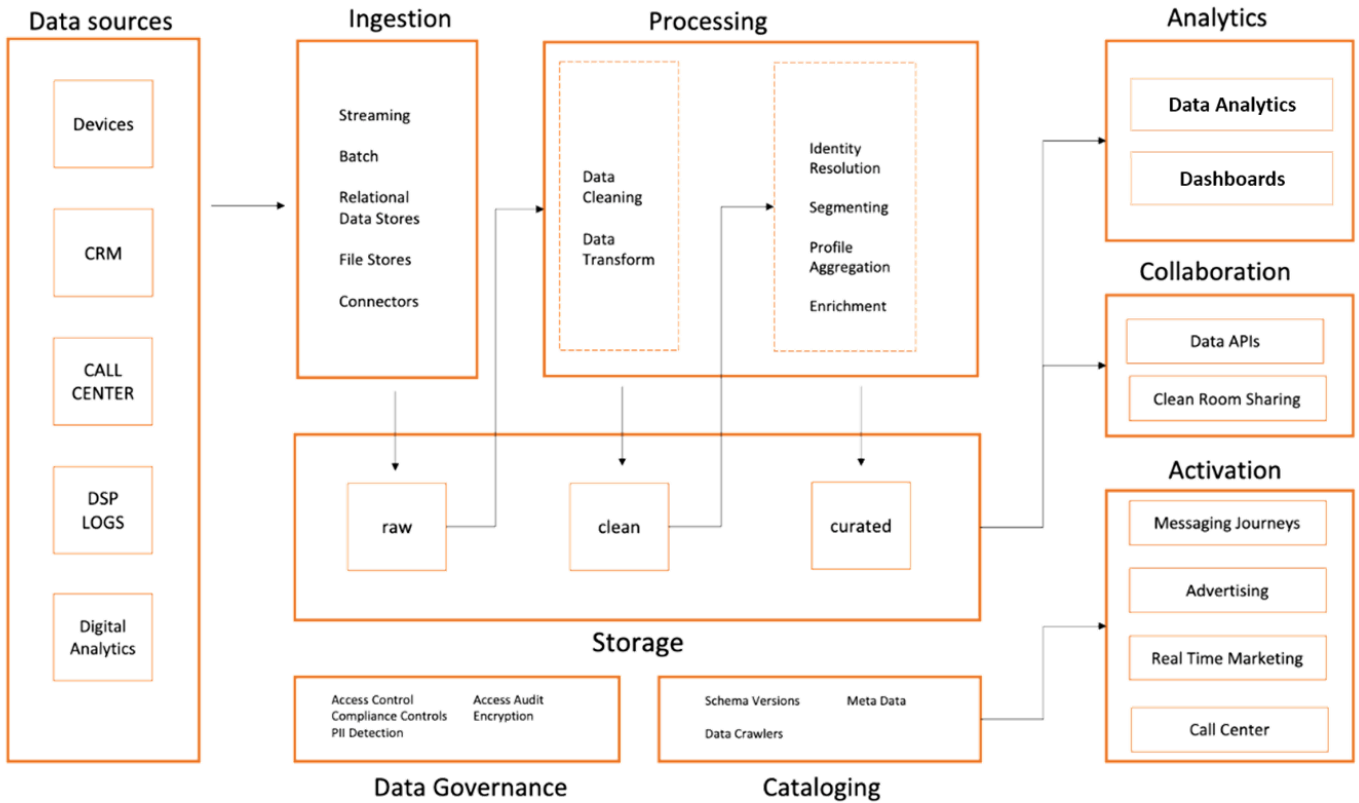
Evolution of data platform

Legacy System	Data warehouse (DWH)	Data lake	Lakehouse / Data Mesh
On-prem databases, Excel reports	*Structured, SQL-based, finance/reporting focus*	*Stores everything (structured + raw) cheaply*	*Unified governance, scalable compute, domain ownership*
Data siloed by department; no real-time access	Slow to adapt, expensive, limited unstructured data	Chaos without governance; “data swamp” risk	Requires culture shift, federated accountability

Data Platform building blocks





Source: https://aws.amazon.com/blogs/architecture/overview-and-architecture-building-customer-data-platform-on-aws/

“The Data Platform is not an IT project — it’s the foundation for the intelligent, decarbonized energy company.”

Conclusion and take away

Data is basic infrastructure

Governance & integration are business enablers, not IT cost

Modern platforms convert data-rich companies into insight-rich companies

Q & A

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